Table S6 Out-of-fold performance of ligand-chemistry models

Target pair	Direction	Common OOF n	Descriptor AUROC	ECFP4 AUROC	ECFP4 + S1 AUROC	Matched S1 AUROC	ECFP4 + S1 increment
EGFR/HER2	Dual vs A-only	65	0.6757	0.8002	0.7838	0.6274	-0.0164
EGFR/HER2	Dual vs B-only	59	0.7592	0.8790	0.8767	0.2408	-0.0023
JAK1/JAK2	Dual vs A-only	64	0.6030	0.9185	0.9199	0.5908	0.0015
JAK1/JAK2	Dual vs B-only	64	0.9014	0.9678	0.9707	0.7485	0.0029
JAK1/TYK2	Dual vs A-only	63	0.5655	0.8488	0.8438	0.6371	-0.0050
JAK1/TYK2	Dual vs B-only	63	0.8407	0.9042	0.9032	0.4032	-0.0010
PIK3CA/mTOR	Dual vs A-only	64	0.6094	0.8213	0.8428	0.7363	0.0215
PIK3CA/mTOR	Dual vs B-only	64	0.2939	0.7285	0.7256	0.5801	-0.0029
AChE/BChE	Dual vs A-only	52	0.6815	0.8993	0.8948	0.6341	-0.0044
AChE/BChE	Dual vs B-only	55	0.7884	0.8201	0.8148	0.6442	-0.0053
F2/F10	Dual vs A-only	63	0.6139	0.9405	0.9335	0.4204	-0.0071
F2/F10	Dual vs B-only	63	0.4889	0.6633	0.6996	0.3740	0.0363
PPARG/PPARA	Dual vs A-only	63	0.7198	0.8206	0.8065	0.6764	-0.0141
PPARG/PPARA	Dual vs B-only	64	0.7158	0.6670	0.6611	0.6021	-0.0059
PPARA/PPARD	Dual vs A-only	64	0.4375	0.9316	0.9287	0.6787	-0.0029
PPARA/PPARD	Dual vs B-only	64	0.3115	0.8740	0.8789	0.5107	0.0049

All models and S1 are evaluated on identical members within each direction. The increment is AUROC(ECFP4 + S1) minus AUROC(ECFP4). Existing frozen five-fold assignments are retained; Q110 uses sorted canonical Murcko scaffolds assigned cyclically to folds 0-4. Standardization is fitted in training folds. Full descriptors, scaffolds, similarities, folds and OOF predictions are supplied as data files; no new model is fitted during table preparation. OOF refers to the learned chemistry models; S1 is a direct docking score.